ApexPetro Refining Complex

Coastal Refinery – Unit 2

P-102 Operating and Inspection Report

Document ID	OPS-P102-2026-08
Asset ID	P-102
Department	Operations / Inspection & Integrity
Report Date	25 August 2026

1. Asset Information

Field	Value
Asset ID	P-102
Service	Feed Transfer Pump
Location	Pump Bay 2 / Unit 2
Motor Power	315 kW
Speed	1485 RPM
Design Flow	475 m3/h

2. Operating Limits

Parameter	Operating Limit	Unit
Discharge Pressure	18 - 22	bar
DE Vibration	<= 4.5	mm/s
Bearing Temperature	<= 85	degC
Flow Rate	450 - 500	m3/h

3. Historical Inspection Table

Date	Parameter	Value	Unit	Status
15 Jan 2025	Discharge Pressure	18.4	bar	Normal
18 Apr 2025	DE Vibration	4.8	mm/s	High
22 Jul 2025	Bearing Temperature	78	degC	Normal
11 Oct 2025	Flow Rate	462	m3/h	Normal
19 Jan 2026	Discharge Pressure	17.2	bar	Low
07 Mar 2026	DE Vibration	5.6 (post-maint. 3.9)	mm/s	High
14 May 2026	Bearing Temperature	82	degC	Normal

Date	Parameter	Value	Unit	Status
21 Aug 2026	Flow Rate	438	m3/h	Low

4. March 2026 Vibration Incident

On 7 March 2026, P-102 DE vibration was recorded at 5.6 mm/s, exceeding the 4.5 mm/s action threshold. A repeat measurement confirmed the elevated reading. A maintenance inspection was requested, and the inspection identified a coupling alignment issue as the cause.

5. Corrective Action

The coupling alignment was corrected using standard shaft alignment procedures. Post-maintenance DE vibration was measured at 3.9 mm/s, within the 4.5 mm/s operating limit. P-102 was returned to service with a monitoring period.

6. Latest-Condition Summary

Parameter	Latest Value	Date	Status
Discharge Pressure	17.2 bar	19 Jan 2026	Low
DE Vibration	3.9 mm/s (post-maint.)	07 Mar 2026	Normal
Bearing Temperature	82 degC	14 May 2026	Normal
Flow Rate	438 m3/h	21 Aug 2026	Low

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